

Misspecification-Robust Inference for the Incremental Fit Indices CFI and TLI under an Estimated Weight

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Purpose

The companion notes fix the estimated-weight inference of a *single* quadratic-form criterion. `rmsea_asymptotics.tex` and the RMSEA path of `higher_order_discrepancy_misspec` treat the discrepancy $F = r'Wr$ itself; the CRMR path treats the correlation residual sum $G = r'V_0r$. Both reduce to one object: a statistic $T = N\hat{F}$, a bias-corrected noncentrality $\hat{\delta} = T - \text{tr}(Q\Gamma_x)$ where Q is a profile Hessian and Γ_x the joint NACOV of the extended first-stage vector $x = (u, \gamma)$, and a sampling variance $\text{Var}(T) = N g'\Gamma_x g$ obtained by the delta method from an envelope-score gradient g .

This note opens the program's *first incremental* (two-model) index. CFI and TLI are not functions of one criterion; they compare the fitted user model to the independence baseline,

$$\text{CFI} = 1 - \frac{\max(0, T_u - df_u)}{\max(0, T_b - df_b)}, \quad \text{TLI} = \frac{T_b/df_b - T_u/df_u}{T_b/df_b - 1}. \quad (1)$$

Under an estimated weight neither T_u nor T_b is χ^2 : each is a mixture $\sum_j \lambda_j \chi_1^2$ whose mean is the generalized degrees of freedom $\bar{Q} = \text{tr}(Q\Gamma_x)$, not the nominal df . The single new statistical object is the *joint* law of (T_u, T_b) . The headline is that both statistics are smooth functionals of the *same* x , so their joint covariance is one bilinear form in the *one* shared Γ_x :

$$\text{Cov}(T_u, T_b) = N g'_u \Gamma_x g_b, \quad (2)$$

and CFI and TLI are then a ratio delta-method on the two correlated noncentralities. No new covariance machinery is needed beyond the second gradient

g_b and the second profile Hessian Q_b ; the metric Γ_x is shared. Three consequences follow: an honest Wald interval for CFI (Corollary 1), a TLI interval that is the CFI interval rescaled by a generalized-df ratio (Corollary 2), and a baseline-dominated simplification (Remark) under which CFI's variance collapses to the RMSEA-side variance the program already computes, divided by the squared baseline noncentrality.

1 Setup

Write $x = (u, \gamma) \in \mathbb{R}^{2m}$ for the extended first-stage vector: the m polychoric sample moments u (thresholds and correlations, stacked in the `OrdinalStats` order) and the DWLS weight diagonal $\gamma = \text{diag } \hat{\Gamma}$, $W = \text{diag}(1/\gamma)$. Let

$$\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x),$$

with Γ_x the joint per-unit NACOV built from the stacked casewise influence $[g_i \mid \text{IF}_i(\gamma)]$ exactly as in `ordinal_dwls_profile_rmsea`. Γ_x is a property of the data, not of any fitted model: *the user model and the baseline share it*.

For a model $M \in \{u, b\}$ (user, baseline) the minimum-discrepancy fit profiles its parameters out of

$$F_M(x) = d_M(x)' W d_M(x), \quad d_M = \sigma_M(\hat{\theta}_M) - u,$$

giving statistic $T_M = NF_M$, profile Hessian Q_M (the $2m \times 2m$ extended object of `weighted_moment_profile_rmsea_estimated_weight`), generalized df $\bar{Q}_M = \text{tr}(Q_M \Gamma_x)$, and bias-corrected noncentrality

$$\hat{\delta}_M = T_M - \bar{Q}_M. \tag{3}$$

The user model is the fitted CFA/SEM. The baseline is the independence model: free thresholds, all polychoric correlations fixed to zero (delta scaling fixes the variances). Its residual is $d_b = (0; -\text{vech}_{<} \hat{R})$ (thresholds profile to zero residual under diagonal W ; the correlation block is the negative sample correlations), and T_b matches `ordinal_baseline_chi2`. Because σ_b is *linear* in the baseline thresholds, the baseline observed bread equals its Gauss–Newton bread and Q_b is exact.

Envelope gradient. Since $\hat{\theta}_M$ minimizes F_M , the $\hat{\theta}_M$ -movement term in dF_M/dx vanishes (envelope theorem) and the gradient is the bare profile

score, identical in form for both models:

$$g_M = \left(\frac{\partial F_M}{\partial u}, \frac{\partial F_M}{\partial \gamma} \right) = \left(-2Wd_M, -d_M^2/\gamma^2 \right) \in \mathbb{R}^{2m}, \quad (4)$$

the γ block coming from $\partial W/\partial \gamma_k = -1/\gamma_k^2$ in $F_M = \sum_k d_{M,k}^2/\gamma_k$. Equation (4) is exactly the RMSEA envelope score, evaluated once at the user residual d_u and once at the baseline residual d_b .

2 Joint law of the two statistics

Theorem 1. *Under fixed misspecification of both models and the regularity of the single-criterion program, with $\hat{x} = \bar{x}$ the sample first-stage vector,*

$$\sqrt{N} \begin{pmatrix} F_u - F_{0,u} \\ F_b - F_{0,b} \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} g'_u \Gamma_x g_u & g'_u \Gamma_x g_b \\ g'_b \Gamma_x g_u & g'_b \Gamma_x g_b \end{pmatrix} \right).$$

Equivalently, with $V_{uu} = Ng'_u \Gamma_x g_u$, $V_{bb} = Ng'_b \Gamma_x g_b$, $V_{ub} = Ng'_u \Gamma_x g_b$, the statistic pair satisfies $\text{Var}(T_u) = V_{uu}$, $\text{Var}(T_b) = V_{bb}$, $\text{Cov}(T_u, T_b) = V_{ub}$, which is (2).

Proof. Each $F_M(\hat{x}) = F_{0,M} + g'_M(\hat{x} - x_0) + o_p(N^{-1/2})$ by a first-order expansion, with g_M the envelope gradient (4) (the profiled parameters contribute no first-order term). Stacking the two expansions and applying the single CLT $\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x)$ to the common $\hat{x}$ gives the bivariate limit; the off-diagonal is $g'_u \Gamma_x g_b$ because both rows read the same increment $\hat{x} - x_0$. Scaling by N ($T_M = NF_M$) gives the stated variances. $\square$

The mean corrections $\bar{Q}_M$ are $O_p(1)$ and, as in the single-criterion program, are treated as fixed in the variance (their sampling fluctuation is higher order). Hence $\text{Var}(\hat{\delta}_u) = V_{uu}$, $\text{Var}(\hat{\delta}_b) = V_{bb}$, $\text{Cov}(\hat{\delta}_u, \hat{\delta}_b) = V_{ub}$.

3 CFI

In the regime where both models misfit, $\hat{\delta}_u > 0$ and $\hat{\delta}_b > 0$, the truncations in (1) are inactive and, writing the generalized-df noncentralities for df ,

$$\widehat{\text{CFI}} = 1 - r, \quad r := \frac{\hat{\delta}_u}{\hat{\delta}_b}. \quad (5)$$

The population target is $\text{CFI}_0 = 1 - \delta_{0,u}/\delta_{0,b} = 1 - F_{0,u}/F_{0,b}$ (the N cancels in the ratio).

Corollary 1. *The ratio gradient is $\nabla r = \hat{\delta}_b^{-1}(1, -r)$, so by the delta method*

$$\widehat{\text{Var}}(\widehat{\text{CFI}}) = \widehat{\text{Var}}(r) = \frac{1}{\hat{\delta}_b^2} \left(V_{uu} - 2r V_{ub} + r^2 V_{bb} \right), \quad (6)$$

and a two-sided $1 - \alpha$ interval is $\widehat{\text{CFI}} \pm z_{1-\alpha/2} \widehat{\text{Var}}(\widehat{\text{CFI}})^{1/2}$, clipped to $[0, 1]$.

4 TLI

TLI is the same comparison taken per generalized degree of freedom. With $E[T_M] \approx \bar{Q}_M + \delta_M$, the misspecification-robust analogue of T/df is $T_M/\bar{Q}_M \approx 1 + \delta_M/\bar{Q}_M$, and substituting into the TLI form of (1) gives

$$\widehat{\text{TLI}} = \frac{T_b/\bar{Q}_b - T_u/\bar{Q}_u}{T_b/\bar{Q}_b - 1} = 1 - cr, \quad c := \frac{\bar{Q}_b}{\bar{Q}_u}. \quad (7)$$

Thus TLI is CFI's ratio r rescaled by the generalized-df ratio c ; $\widehat{\text{TLI}} = 1 - (1 - \widehat{\text{CFI}}) \bar{Q}_b/\bar{Q}_u$. Using $\bar{Q}$ rather than the nominal df keeps TLI internally consistent with the mixture mean that defines $\hat{\delta}$; when the weight is correctly specified $\bar{Q}_M \rightarrow df_M$ and (7) recovers the lavaan point value.

Corollary 2. *Treating c as fixed (the $\bar{Q}_M$ are deterministic mean corrections),*

$$\widehat{\text{TLI}} = 1 - cr \implies \widehat{\text{Var}}(\widehat{\text{TLI}}) = c^2 \widehat{\text{Var}}(\widehat{\text{CFI}}). \quad (8)$$

The TLI interval is the CFI interval scaled by $c = \bar{Q}_b/\bar{Q}_u$: same r , same $\widehat{\text{Var}}(r)$, one extra factor. (TLI is not clipped to $[0, 1]$; it can exceed 1 for $\hat{\delta}_u < 0$.)

5 The baseline-dominated regime

Under any appreciable misspecification the independence model fits grossly, so $\hat{\delta}_b \gg \hat{\delta}_u$, $r \rightarrow 0$, and (6) collapses to its leading term:

$$\widehat{\text{Var}}(\widehat{\text{CFI}}) = \frac{V_{uu}}{\hat{\delta}_b^2} \left(1 - 2r \frac{V_{ub}}{V_{uu}} + r^2 \frac{V_{bb}}{V_{uu}} \right) \approx \frac{V_{uu}}{\hat{\delta}_b^2} = \frac{\text{Var}(T_u)}{\hat{\delta}_b^2}. \quad (9)$$

That is, to leading order CFI's sampling variance is the *user-model* statistic variance $V_{uu} = Ng'_u \Gamma_x g_u$ already returned as `grad_var` by the RMSEA path, divided by the squared baseline noncentrality $\hat{\delta}_b^2$; the baseline acts as a nearly constant scale. The cross-term and baseline-variance contributions are $O(r)$ and $O(r^2)$ corrections. The full bivariate form (6) is retained in code for honesty, but (9) is the right intuition and an effective cross-check: *CFI inference is a rescaling of RMSEA inference.*

6 Estimated versus fixed weight

Zeroing the γ blocks of Γ_x (drop the $\text{IF}(\gamma)$ rows/columns) gives the fixed-weight comparator: the g_M γ -channel in (4) is then inert, V_{MM} and V_{ub} revert to their conventional (treat- W -as-known) values, and $\bar{Q}_M$ reverts to the fixed-weight trace. The estimated-weight law differs from current practice exactly through this channel. It enters CFI twice: through V_{uu} (the RMSEA-metric piece, where the channel is large because RMSEA's metric *is* the weight) and through the baseline objects $\hat{\delta}_b, V_{bb}$. By (9) the dominant effect is the former, so the estimated-weight correction to CFI tracks the estimated-weight correction to the user RMSEA, attenuated by $1/\hat{\delta}_b^2$.

7 Boundary and regularity

- **User boundary.** As $\hat{\delta}_u \downarrow 0$ (correct or near-correct user model) $\widehat{\text{CFI}} \rightarrow 1$ sits on the parameter-space boundary, the truncation $\max(0, \cdot)$ activates, and the normal approximation degrades. The interval is clipped at 1 and flagged. This is the RMSEA close-fit boundary made sharper by CFI's hard ceiling.
- **Baseline boundary.** If $\hat{\delta}_b \leq 0$ (independence model fits; essentially uncorrelated data) the ratio is undefined: report $\widehat{\text{CFI}} = 1$, $\widehat{\text{TLI}}$ and both intervals NaN, with a warning. The delta method requires $\hat{\delta}_b$ bounded away from zero, which holds whenever the baseline is a genuine null.
- **Shared metric.** $g'_u \Gamma_x g_b$ uses the *same* Γ_x for both gradients; the implementation must assert the two profile fits returned a common Γ_x (they do, being data-only) before forming the cross-term.
- **Single group.** As with RMSEA/CRMR, the first pass is single-group; the multi-group extension stacks Γ_x block-diagonally and sums the block traces, and is deferred.

8 Implementation map

1. User profile via `ordinal_dwls_profile_rmsea` $\Rightarrow Q_u, \Gamma_x, T_u, \bar{Q}_u$; form g_u from (4) with the fitted residual d_u (as the RMSEA path already does).

2. Baseline profile from the analytic independence block: jacobian $D_b = [I_{nth}; 0]$, residual d_b , weight γ , shared Γ_x , Gauss–Newton bread $D'_b W D_b$, $K_b = I$. Route through `weighted_moment_profile_rmsea_estimated_weight` $\Rightarrow Q_b, T_b, \bar{Q}_b$; cross-check T_b against `ordinal_baseline_chi2`. Form g_b from (4) with d_b .
3. $\hat{\delta}_u = T_u - \bar{Q}_u$, $\hat{\delta}_b = T_b - \bar{Q}_b$; $V_{uu} = Ng'_u \Gamma_x g_u$, $V_{bb} = Ng'_b \Gamma_x g_b$, $V_{ub} = Ng'_u \Gamma_x g_b$.
4. CFI (5), (6); TLI (7), (8); intervals per Corollaries 1–2 with the boundary handling of Section 7.
5. `estimated_weight=false` zeros the γ blocks of Γ_x before steps 3–4 (fixed-weight comparator).